# UNDERGRADUATE RESEARCH SUMMARY

## *Sibling Network Size and Household Debt Ratio in China: An Empirical Analysis Based on the CHFS 2017 Survey*

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### 1. Introduction and Research Question

Household indebtedness in China has surged since the early 2000s, raising concerns about financial stability and household welfare (Liu et al., 2020; Tseng & Hsiao, 2022). This thesis investigates the relationship between an under-explored dimension of family structure—the sibling network of the household head—and the household debt-to-asset ratio. The study is situated at the intersection of family economics, social capital theory (Bourdieu, 1986; Coleman, 1988), and behavioural household finance. Four theoretical channels are identified: (i) parental resource dilution (Bai, 2021), which may constrain borrowing capacity; (ii) intra-sibling risk sharing, increasing tolerance for financial risk (Wu & Zhao, 2020); (iii) life-event expenditure pressure amplified by familial obligations (Zhang, 2024); and (iv) peer effects and social comparison among siblings.

The central research question is: does a systematic correlation exist between the number of siblings of the household head and the household debt ratio in China? We formulate:

> **$H_1$:** *Controlling for other relevant factors, there exists a positive and statistically significant correlation between the number of siblings of the household head and the household debt ratio.*

### 2. Data and Sample Construction

The analysis draws on the China Household Finance Survey (CHFS) 2017, a nationally representative survey conducted by Southwestern University of Finance and Economics (SWUFE), covering approximately 40,000 households across 29 provinces under a multi-stage stratified probability sampling design. Individual- and household-level datasets were merged; the household head was identified via the relationship variable (a2001 = 1). The sibling count was computed by summing brothers (a2028) and sisters (a2029), items administered only to respondents aged ≤ 40, restricting the sample to the 16–40 age range. Listwise deletion of missing values yields a final estimation sample of N = 7,776 households.

For asset and liability components recorded as bracketed intervals, a midpoint estimation function was implemented, assigning to each interval code the central value of the corresponding range defined in the survey instrument.

### 3. Econometric Specification

The dependent variable is the household debt ratio, defined as:

$$R_i = D_i \,/\, (A_i + \varepsilon) \qquad (1)$$

where $D_i$ denotes total liabilities, $A_i$ total assets of household i, and $\varepsilon = 10^{-9}$ is a small constant preventing division by zero. This variable is winsorised at the 1st and 99th percentiles ($R_i^W$). The baseline OLS model is:

$$R_i^W = \beta_0 + \beta_1 \cdot Siblings_i + X_i'\Gamma + \varepsilon_i \qquad (2)$$

where $Siblings_i$ is the sibling count of household head i; $X_i$ is a vector of controls (age, sex, education, marital status, self-rated health, business ownership, number of properties, log total assets); $\beta_1$ is the coefficient of interest; $\Gamma$ the control coefficient vector. Three robustness checks complement the analysis: (a) OLS with log-transformed dependent variable $\log(R_i^W + 0.001)$; (b) Ridge Regression (RidgeCV with cross-validation), although all VIFs < 1.7; (c) Robust Linear Model (RLM) with Huber's T norm, addressing non-normality confirmed by the Omnibus and Jarque–Bera tests ($p < 0.001$).

## 4. Results

Table 1 reports the estimated coefficient $\hat{\beta}_1$ on Siblings across the five specifications.

**Table 1 — Estimates of $\beta_1$ (Siblings)**

| Model | Method | Dep. Var. | $\hat{\beta}_1$ | Std. Err. | p-value |
|---|---|---|---|---|---|
| (1) | OLS | $R^W$ | 0.0483 | 0.013 | < 0.001 *** |
| (2) | OLS | $\log(R^W + 0.001)$ | 0.1794 | 0.024 | < 0.001 *** |
| (3) | Ridge | $R^W$ | $0.067^s$ | — | — |
| (4) | Ridge | $\log(R^W + 0.001)$ | $0.247^s$ | — | — |
| (5) | RLM (Huber) | $R^W$ | 0.0051 | 0.001 | < 0.001 *** |

*Notes: *** $p < 0.01$. $^s$ Standardised coefficients (Ridge). N = 7,776. Durbin–Watson ≈ 2.00. VIF < 1.7 for all regressors. Adj. $R^2$: 0.076 (Model 1), 0.040 (Model 2).*

The coefficient $\hat{\beta}_1$ is positive and highly significant ($p < 0.001$) across all three parametric models (OLS levels, OLS log, RLM) and retains a positive sign under Ridge. Under Model (1), ceteris paribus, each additional sibling is associated with a 4.83 p.p. increase in the winsorised debt ratio. Controls exhibit expected signs: positive effects for marital status, business ownership, and property count; negative wealth effect on the debt ratio (OLS).

## 5. Discussion and Conclusion

The convergence of results across five specifications suggests the sibling network is a significant correlate of household indebtedness in China. Three mechanisms appear predominant: intra-family risk sharing raising debt tolerance (Wu & Zhao, 2020); financial pressure from cultural and life-event obligations amplified by sibling network size (Zhang, 2024); and peer effects fostering social comparison and imitation of borrowing behaviour.

Limitations include the correlational nature of findings (no causal identification), cross-sectional data, and low $R^2$ reflecting multifactorial complexity. Future research directions include panel data, instrumental variable strategies (e.g., regional family planning policy variation), and disaggregation by debt type.

## Selected References


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